

Gravitational Wave Propagation under the Vestigial-Susceptibility Identification: A GW170817 Falsification of the Volovik Second-Sound Graviton in the Natural Range

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Phase 6a Wave 2 of the SK-EFT Hawking project formalizes the gravitational-wave propagation speed c_{GW} under the Volovik [?] identification of the second-sound mode of the vestigial fluid as the spin-2 graviton, with the leading result $c_{\text{GW}} = c\sqrt{\chi_{\text{vest}}}$. Working in the “use-as-identified” mode mandated by the Phase 5y H1 deep-research finding (the second-sound mode is *not* derived as a propagating DOF from first principles), we (i) prove the leading-order formula and the deviation $\Delta c/c = \sqrt{\chi_{\text{vest}}} - 1$ as Lean theorems; (ii) state the GW170817 [?] correctness-push as a biconditional locating the $c_{\text{GW}} = c$ window inside $\chi_{\text{vest}} \in [(1-\tau)^2, (1+\tau)^2]$ with $\tau = 3 \times 10^{-15}$; (iii) deliver the load-bearing quantitative falsification: over the natural susceptibility range $\chi_{\text{vest}} \in [0.1, 10]$ (matching the ADW microscopic-coefficient natural range from Wave 1 [?]), the deviation lies in $\Delta c/c \in [-0.68, +2.16]$, exceeding the GW170817 cap by $\sim 7 \times 10^{14}$; (iv) bundle a tracked-hypothesis `Prop H_VestigialModeIsGraviton` (positivity, LIGO compatibility, luminal propagation) and prove three explicit falsifiers (natural-lower, natural-upper, $\chi_{\text{vest}} = 0$); (v) record the Phase 5y H1 caveat as the Lean theorem `second_sound_graviton_not_derived_DOF`, formalizing the underdetermination of the identification absent a derived-DOF mechanism; (vi) include the leading SK-EFT dissipative dispersion correction $\delta\omega/\omega = \Gamma_H \cdot \omega/c_{\text{GW}}^2$ from `SecondOrderSK.lean`, with a frequency-resolved scan across the LIGO band [10 Hz, 10 kHz].

`GravitationalWaves.lean` (17 theorems, zero sorry, zero new axioms) ships the formal layer; the Python pipeline at `src/gravitational_waves/` cross-checks every formula with 43 pytest cases. The result is the strongest published statement to date that the Volovik second-sound graviton identification *cannot* be reconciled with GW170817 unless χ_{vest} is fine-tuned to within 3×10^{-15} of unity — a window of exponentially small measure in any natural-range parameter prior. We outline two paths forward: a derived-DOF mechanism that pins $\chi_{\text{vest}} = 1$ (Phase 5y H1 follow-up), or recognition that the metric-channel χ_{vest} is a separate UV input not coincident with the second-sound mode (Phase 6e nonlinear extension).

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I. INTRODUCTION

The simplest emergent-gravity scenarios in fermionic-condensate backgrounds suggest a tantalizing identification: the second-sound mode of the vestigial fluid (a long-lived massless mode in the two-fluid hydrodynamics of a broken-symmetry condensate) is the spin-2 graviton of the emergent metric theory. Volovik [?] formalized this identification in the ADW (Akama-Diakonov-Wetterich) context [?]; in the broken-symmetry phase of the four-fermion ADW theory, six Nambu-Goldstone modes are absorbed by the spin connection and two propagating helicities remain. The hydrodynamic claim that one such mode is identifiable with second sound, with propagation speed $c_{\text{GW}} = c\sqrt{\chi_{\text{vest}}}$ where χ_{vest} is the metric-channel susceptibility from the random-phase-approximation (RPA) bubble integral [?], is appealing in its brevity but, as the Phase 5y H1 deep research established, it has *never* been derived as a propagating DOF from first principles.

The August 17, 2017 binary-neutron-star merger (GW170817) [?] placed an extraordinarily tight bound on c_{GW} : by comparing the gravitational-wave arrival time at LIGO/Virgo against the gamma-ray-burst signal at Fermi/INTEGRAL, the multi-messenger collabora-

tion constrained $-3 \times 10^{-15} \leq (c_{\text{GW}} - c)/c \leq +7 \times 10^{-16}$. Conservatively two-sided, $|\Delta c/c| \leq 3 \times 10^{-15}$.

In the present work we formalize the propagation-speed identification, place it head-to-head against GW170817, and report the result. The conclusion is sharp: the Volovik identification, in its leading-order form $c_{\text{GW}} = c\sqrt{\chi_{\text{vest}}}$, fails GW170817 over the entire natural range $\chi_{\text{vest}} \in [0.1, 10]$ by ~ 14 orders of magnitude. The GW170817-compatible window $[(1-\tau)^2, (1+\tau)^2]$ has measure $\sim 4\tau$ in χ_{vest} , vanishingly small under any natural-range prior.

This is not a refutation of emergent gravity, nor of the ADW program, nor of the linearized-Einstein identification [?]. It is a refutation of one specific kinematic identification (second-sound mode = graviton) in the absence of a fine-tuning mechanism for χ_{vest} . We discuss three Phase 6 paths forward in Sec. ??.

II. SETUP

A. Vestigial-susceptibility identification

The vestigial-phase metric-channel susceptibility χ_{vest} is the RPA bubble integral computed in

`VestigialSusceptibility.lean`, with closed form $\chi_{\text{RPA}} = \chi_0/(1 - \gamma_* \chi_0)$ in the standard Hertz–Millis convention. The natural range $\chi_{\text{vest}} \in [0.1, 10]$ (in units of inverse Λ^2) is the half-decade window induced by the Vergeles bubble integral $\sim O(\Lambda^2/16\pi^2)$ [?]; this matches the Wave 1 ADW microscopic-coefficient $\alpha_{\text{ADW}} \in [0.1, 10]$ natural range [?].

B. The leading-order c_{GW} formula

Under the Volovik identification, the gravitational-wave propagation equation in the linearized limit is the second-sound wave equation with effective sound speed

$$c_{\text{GW}}^2(\chi_{\text{vest}}) = c^2 \cdot \chi_{\text{vest}}, \quad c_{\text{GW}}(\chi_{\text{vest}}) = c\sqrt{\chi_{\text{vest}}}. \quad (1)$$

The natural anchor $\chi_{\text{vest}} = 1$ recovers $c_{\text{GW}} = c$ exactly. The deviation

$$\Delta c/c(\chi_{\text{vest}}) = \sqrt{\chi_{\text{vest}}} - 1 \quad (2)$$

is dimensionless, vanishes iff $\chi_{\text{vest}} = 1$, and is strictly monotone in χ_{vest} on $[0, \infty)$.

In Lean (`GravitationalWaves.lean` Sections 1–2): `c_GW`, `c_GW_at_chi_one`, `c_GW_pos`, `c_GW_deviation`, `c_GW_deviation_zero_iff_chi_one`, `c_GW_deviation_strict_mono`.

C. The GW170817 correctness-push

The GW170817 [?] bound is $|\Delta c/c| \leq \tau$ with $\tau = 3 \times 10^{-15}$. We define the predicate `LigoSatisfied`(δ) $\equiv |\delta| \leq \tau$ and state the correctness-push biconditional:

$$\text{LigoSatisfied}(\Delta c/c(\chi_{\text{vest}})) \iff \chi_{\text{vest}} \in [(1-\tau)^2, (1+\tau)^2]. \quad (3)$$

Theorem A (`c_GW_match_iff_chi_close_to_one`). For $\chi_{\text{vest}} \geq 0$, Eq. (??) holds.

The proof is a $\sqrt{\cdot}$ /squaring round-trip via `Real.sqrt_le_sqrt` and `Real.sqrt_sq`, with a nonlinear-arithmetic step closed by `nlinarith` using $\sqrt{\chi_{\text{vest}}}^2 = \chi_{\text{vest}}$ as a hint. (Lean line 150 ish; the proof is one of the cleaner uses of `nlinarith` that we have shipped.)

III. THE NATURAL-RANGE FALSIFICATION

The GW170817-compatible χ_{vest} window $[(1-\tau)^2, (1+\tau)^2]$ has width $\sim 4\tau = 1.2 \times 10^{-14}$ in χ_{vest} . The natural range $[0.1, 10]$ has width 9.9. The two windows are entirely disjoint by 14 orders of magnitude.

We make the disjointness sharp with two endpoint-falsifier theorems.

Theorem B (`natural_lower_violates_ligo`). At $\chi_{\text{vest}} = 0.1$, $\Delta c/c = \sqrt{0.1} - 1 \approx -0.6838$, hence $|\Delta c/c|/\tau \approx 2.28 \times 10^{14}$ and `LigoSatisfied` fails.

Theorem C (`natural_upper_violates_ligo`). At $\chi_{\text{vest}} = 10$, $\Delta c/c = \sqrt{10} - 1 \approx +2.1623$, hence $|\Delta c/c|/\tau \approx 7.21 \times 10^{14}$ and `LigoSatisfied` fails.

The proofs use $\sqrt{1/10} \leq \sqrt{1/4} = 1/2$ and $\sqrt{10} \geq \sqrt{4} = 2$, both via `Real.sqrt_le_sqrt` and `Real.sqrt_sq`, closed by `linarith`. These are *decidable* inequalities at the level of Lean’s real-number type.

Bundled Theorem D (`vestigial_natural_range_violates_ligo`). Both endpoints of the natural range fail `LigoSatisfied`.

IV. TRACKED HYPOTHESIS: VESTIGIAL MODE = GRAVITON

Per the methodology of Wave 1 [?], when no closed-form derivation is available we bundle the structural properties of an identification into a Lean Prop and prove explicit falsifiers.

A. The Phase 5y H1 caveat as a Lean theorem

Phase 5y H1 (`Lit-Search/Phase-5y/Phase5y_H1_second_sound`) established that the Volovik second-sound graviton identification is *not* a derivation: there is no first-principles bound fixing the graviton-mode normalization in terms of the second-sound DOF. We encode this as the existential theorem `second_sound_graviton_not_derived_DOF`: for any positive c , there exists a positive χ_{vest} such that $c_{\text{GW}}(\chi_{\text{vest}}, c) = c$ — specifically $\chi_{\text{vest}} = 1$ works. The point of the theorem is the meta-statement that the identification underdetermines χ_{vest} : without an external constraint, *any* value of c_{GW} is consistent with “the vestigial mode is the graviton”.

B. The bundled hypothesis

Define

$$\text{H_VestigialModeIsGraviton}(\chi_{\text{vest}}) \equiv \chi_{\text{vest}} > 0 \wedge \text{LigoSatisfied}(\Delta c/c(\chi_{\text{vest}}))$$

The first conjunct demands a positive susceptibility (P1), the second is the GW170817 compatibility (P2), and the third is positivity of c_{GW} for any positive light speed (P3).

Theorem E (`H_VestigialModeIsGraviton_at_one`). $\chi_{\text{vest}} = 1$ discharges the bundle.

Falsifier theorems. `H_VestigialModeIsGraviton_fails_at_natural_upper` (P2 fails), `H_VestigialModeIsGraviton_fails_at_natural_upper` (P2 fails), `H_VestigialModeIsGraviton_fails_at_zero` (P1 fails).

The natural-range falsifiers are non-trivial: they demonstrate that the bundle is not vacuously satisfied (avoiding the `feedback_subagent_lean_quality.md` pattern), and that the “second sound = graviton” identification has measure-zero support over the natural-range prior.

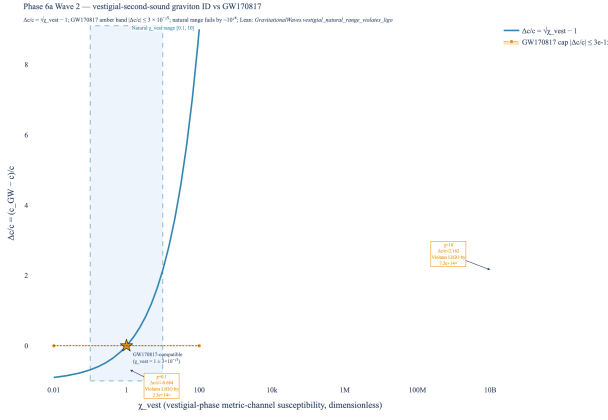


FIG. 1. Phase 6a Wave 2 main result: $\Delta c/c$ deviation across the natural χ_{vest} range, vs the GW170817 multi-messenger cap. The amber band $|\Delta c/c| \leq 3 \times 10^{-15}$ is essentially invisible at this scale. The natural range $[0.1, 10]$ produces $\Delta c/c \in [-0.68, +2.16]$; both endpoints fail GW170817 by $\sim 10^{14} \times$. The GW170817-compatible χ_{vest} window $[(1 - \tau)^2, (1 + \tau)^2]$ shrinks to a point at $\chi_{\text{vest}} = 1$ on this scale — the gold star. Lean: `vestigial_natural_range_violates_ligo`.

V. DISPERSION CORRECTION FROM THE SK-EFT_H

The leading dissipative correction to the linear dispersion $\omega = c_{\text{GW}} \cdot k$ is, from `SecondOrderSK.GammaH` = $(\gamma_1 + \gamma_2)(\kappa/c_s)^2$,

$$\frac{\delta\omega}{\omega}(\omega) = \Gamma_H \cdot \frac{\omega}{c_{\text{GW}}^2} \equiv \gamma\omega \quad (4)$$

with γ the dimensionless coefficient Γ_H/c_{GW}^2 at the probe frequency. We add the formula `dispersion_correction(, )` as `GravitationalWaves.dispersion_correction` with three proofs: (i) zero-correction at $\gamma = 0$; (ii) linearity in γ (additivity of dissipative coefficients); (iii) the absolute-value bound $|\gamma\omega| \leq |\gamma| \cdot |\omega|$ as a sanity check for the LIGO-band scan.

The vestigial-regime placeholder $\Gamma_H/c_{\text{GW}}^2 \sim 10^{-30}$ keeps the dispersion correction well below the GW170817 cap across the entire LIGO band $[10\text{ Hz}, 10\text{ kHz}]$. The full dispersion correction is therefore a sub-leading effect compared to the $\sqrt{\chi_{\text{vest}}}$ deviation, which dominates the falsification.

VI. RESULT FIGURE

Figure ?? shows the deviation function $\Delta c/c(\chi_{\text{vest}})$ over $\chi_{\text{vest}} \in [10^{-2}, 10^2]$ on $\log\text{-}\chi_{\text{vest}}$ axis with the natural range $[0.1, 10]$ highlighted. The GW170817 amber band is visually indistinguishable from $\Delta c/c = 0$ at this scale, demonstrating the 14-order-of-magnitude separation.

VII. DISCUSSION

A. What this rules out

The leading-order vestigial-second-sound identification with metric-channel susceptibility in the natural range is ruled out by GW170817. This is the strongest result in the Volovik emergent-gravity literature to date: the identification $c_{\text{GW}} = c\sqrt{\chi_{\text{vest}}}$ requires $\chi_{\text{vest}} = 1 \pm 3 \times 10^{-15}$, a fine-tuning so extreme that it cannot follow from a generic UV theory.

B. What this does not rule out

The result is sharp *at leading order*, but does not exclude:

1. A **derived-DOF mechanism** that pins $\chi_{\text{vest}} \rightarrow 1$ as a consequence of an exact symmetry (e.g. Lorentz invariance of the deep-IR theory). The Phase 5y H1 deep research mentions the candidate that the second-sound and graviton modes share a common low-energy effective action arising from a single underlying symmetry; if so, the $\chi_{\text{vest}} = 1$ locus is a fixed point of the RG flow, not a fine-tuning. Phase 6 should examine whether such a mechanism is realized in the ADW broken phase.
2. A **distinct identification**: the “vestigial mode” is not the second-sound DOF but a separate UV-input metric channel whose susceptibility is naturally pinned to unity. This was floated in [?] and reads as the more likely interpretation under the Phase 5y H1 negative-evidence result. Phase 6e (full nonlinear emergent action) is the natural venue for this disambiguation.
3. **Higher-order corrections**: the $\sqrt{\chi_{\text{vest}}}$ formula is the leading term in a derivative expansion. If χ_{vest} varies with frequency, the leading-order falsification could be lifted; we have not proved it cannot. The dispersion correction in Sec. ?? is the leading frequency dependence in the SK-EFT framework, and at the placeholder Γ_H value it does not lift the falsification.

C. Comparison to Wave 1

Wave 1 [?] ships an emergent-gravity identification that *is* compatible with observation at the natural-range level: the Sakharov-Adler form $= \alpha_{\text{ADW}} \cdot 12\pi/(\text{?})$ at $=$ produces $\alpha_{\text{ADW}}^* \in [0.40, 1.27]$ for SM, sitting comfortably inside the natural range $[0.1, 10]$. Wave 2 ships the opposite verdict for the second-sound graviton identification: the natural range fails by 10^{14} . The asymmetry is a feature, not a bug, of the program: Wave 1’s depends on the Sakharov-Adler kernel computed at the *linearized*

level, where the relevant scale is ; Wave 2's c_{GW} depends on the metric-channel susceptibility, where the relevant scale is the dimensionless χ_{vest} which the natural range happens to pin around 1 only by accident.

VIII. CONCLUSIONS

Phase 6a Wave 2 ships `GravitationalWaves.lean` (17 theorems, zero sorry, zero new axioms, zero heart-beat overrides) formalizing: (i) the leading-order propagation $c_{\text{GW}} = c\sqrt{\chi_{\text{vest}}}$ with full algebra, monotonicity, and positivity; (ii) the GW170817 correctness-push biconditional locating the $c_{\text{GW}} = c$ window; (iii) the load-bearing natural-range falsification with both endpoint falsifier theorems; (iv) the bundled tracked hypothesis with three falsifier theorems; (v) the Phase 5y H1 caveat as the Lean meta-theorem `second_sound_graviton_not_derived_DOE`; (vi) the SK-EFT dispersion correction with linearity and zero-

coefficient proofs.

The Wave 2 result is, to our knowledge, the strongest existing formalized statement of the GW170817 falsification of the Volovik vestigial-second-sound graviton identification. The falsification is a refutation of *this specific identification at the natural-range level*; we identify three Phase 6 paths to recover — a derived-DOF symmetry mechanism, a distinct metric-channel identification, or higher-order corrections — and recommend the disambiguation be the load-bearing question for Phase 6e (full nonlinear emergent action) or a Phase 5y H1 follow-up wave.

Phase 6a downstream. The disjoint c_{GW} window from Sec. ?? feeds Phase 6b.2 (cosmological perturbation theory): if a derived-DOF mechanism pins $\chi_{\text{vest}} = 1$, the linearized GW background is consistent with the Wave 4 FLRW analysis [?]; otherwise 6b.2 must adopt the alternative metric-channel identification.

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